# Identification Chart for the METAMORPHIC ROCK COLLECTION

*Ward's Natural Science 45 E 3120 · Format adapted from Bob Leighty's Identification Chart for Common Metamorphic Rocks*

| TEXTURE | GRAIN SIZE | ROCK NAME | DESCRIPTION | INTERPRETATIONS | | | ADDITIONAL INFORMATION (setting & parent rock) |
|---|---|---|---|---|---|---|---|
| | | | | Common Protoliths (original or "parent" rock) | Metamorphic Grade / Type | | |
| **Foliated** (layered) | very fine | **RED SLATE** | dull, smooth, rings when struck; layers break into thin flat plates (slaty cleavage); **red-brown** from hematite | red shale, red clay | HIGH ↓ metamorphic grade ↑ LOW | low pressure (P) & low temperature (T) metamorphism (regional) | Shale deposited in a basin is buried and squeezed at a convergent margin (about 200 to 300 °C). Red color: hematite ($Fe^{3+}$) in an oxidized sediment. |
| | very fine | **GRAY SLATE** | same slaty cleavage; **gray to black** from organic carbon in the original mud | black mud, shale with organic matter | | low P & low T metamorphism (regional) | Same setting as red slate. Carbon from buried organic matter, not oxidized, gives the dark gray to black color. |
| | medium to coarse | **SCHIST** | layers of flaky, glistening **mica** or green chlorite (schistosity); no obvious extra minerals | mudstone, shale | | low to moderate P & T metamorphism (regional) | Mudstone heated further in a mountain belt (about 350 to 500 °C); mica grows large enough to see. Chlorite marks lower grade, muscovite and biotite higher. |
| | medium to coarse | **GARNETIFEROUS MICA SCHIST** | mica schist with numerous **garnet** crystals (index mineral) | mudstone, shale | | moderate P & T metamorphism (regional) | Garnet grows at moderate temperature (about 450 to 550 °C and up) in the zone of regional metamorphism in a collision belt. An index mineral for grade. |
| | medium to coarse | **HORNBLENDE SCHIST** | dark green to black; aligned **hornblende** (amphibole) crystals; minerals completely reformed and rearranged | basalt, gabbro (dark igneous rock) | | moderate to high P & T metamorphism (regional); amphibolite facies | Basalt or gabbro (ocean crust) caught in a mountain belt; recrystallizes at about 500 to 700 °C into amphibole plus feldspar (amphibolite facies). |
| | coarse | **BIOTITE GNEISS** | coarse light and dark **compositional bands** (gneissic banding); dark **biotite** streaks; crushed and squeezed | granite (plutonic rock) | | high P & high T metamorphism (regional) | Granite buried deep in a collision zone (about 600 °C or more); minerals segregate into bands. Gneiss from granite is an orthogneiss. Next step up: migmatite. |
| **Non-Foliated** (massive or no layering) | fine to coarse | **QUARTZITE** | interlocking quartz breaks across grains; **very hard (H = 7)**, scratches glass; will not react with HCl; original sand grains hard to distinguish | quartz sandstone (quartz arenite) | variable metamorphic grade; low P & high T (contact) or regional (no foliation: no platy minerals) | | Quartz sandstone recrystallized; quartz grains grow together so the rock breaks through grains, not around them. Forms near intrusions or in mountain belts. |
| | coarse | **GRANITOID GNEISS** | coarse, interlocking quartz, feldspar and dark specks; granite-like; little or **no banding**; scratches glass; no HCl reaction | granite (plutonic rock) | high P & high T metamorphism (regional); foliation weak or absent | | Same granite parent and deep-crust setting as biotite gneiss, but the minerals did not segregate into bands. Gneiss can be nonfoliated. |
| | medium to coarse | **WHITE MARBLE** | white, sugary interlocking **calcite**; easily scratched with a steel nail (H = 3); **fizzes in dilute HCl** | limestone | variable metamorphic grade; contact or regional | | Limestone recrystallizes: calcite grains grow and interlock, destroying fossils and bedding. Forms around intrusions (contact aureoles) or in mountain belts. |
| | medium to coarse | **PINK MARBLE** | pink from impurities; fizzes if calcitic; if **dolomitic**, powder the surface before it reacts | limestone, dolostone | variable metamorphic grade; contact or regional | | Same origin as white marble. Pink color from iron or manganese impurities. If the parent was dolostone, the rock is dolomitic and reacts weakly to acid. |
| | fine | **SERPENTINITE** | dark green, **waxy or greasy** luster; harder than a fingernail, softer than glass; no HCl reaction | peridotite (ultramafic rock) | low metamorphic grade; hydration of olivine and pyroxene to serpentine | | Mantle peridotite reacts with water at low temperature (below about 500 °C) at mid-ocean ridges and subduction zones: olivine becomes serpentine. Found in ophiolites. |
| | fine | **SOAPSTONE** | soft (H = 1), **slippery, soapy feel**; mostly talc; scratched by a fingernail; may be weakly foliated | ultramafic rock altered by Mg-rich fluids | low to moderate grade; low P & T | | Mg-rich ultramafic rock altered by hot, $CO_2$-bearing fluids; talc forms (about 300 to 500 °C). Soft and heat resistant, so used for carving and stoves. |

**Tests:** foliation (rotate under a light) · acid (one drop dilute HCl, goggles on) · hardness: fingernail 2.5, copper coin about 3.5, steel nail and glass about 5.5, quartz 7.

**Grade sequence:** slate, phyllite, schist, gneiss with rising T and P. Phyllite (fine, satiny sheen, wavy surfaces) is not in this collection. Name schists by prominent minerals (garnet, hornblende, etc.).